TST Jurisprudence Explorer

An interactive application to navigate a large legal corpus with embeddings and clustering

CSDS-352 - Data Science and Machine Learning - Capstone Final

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Agenda

1. The problem and the domain
2. The dataset
3. **Key concepts** (embeddings, vector search, UMAP, HDBSCAN, quality ML)
4. System architecture
5. Live demo (dashboard + CLI)
6. Results
7. Engineering: tests, CI, Docker
8. Limitations, future work, references

The problem

- The **Tribunal Superior do Trabalho (TST)** publishes a book of **Súmulas**, **Orientações Jurisprudenciais (OJs)** and **Precedentes Normativos (PNs)**.
- It ships as a dense **579-page PDF** in Portuguese.
- Each statement is mixed with its **historical redactions**, cross-references and cancellation notes.
- Finding related rulings by hand is slow, and there is no machine-readable version to search by meaning.

One sentence: we turn this legal PDF into a searchable, clustered and quality-checked knowledge base.

Domain and requirements

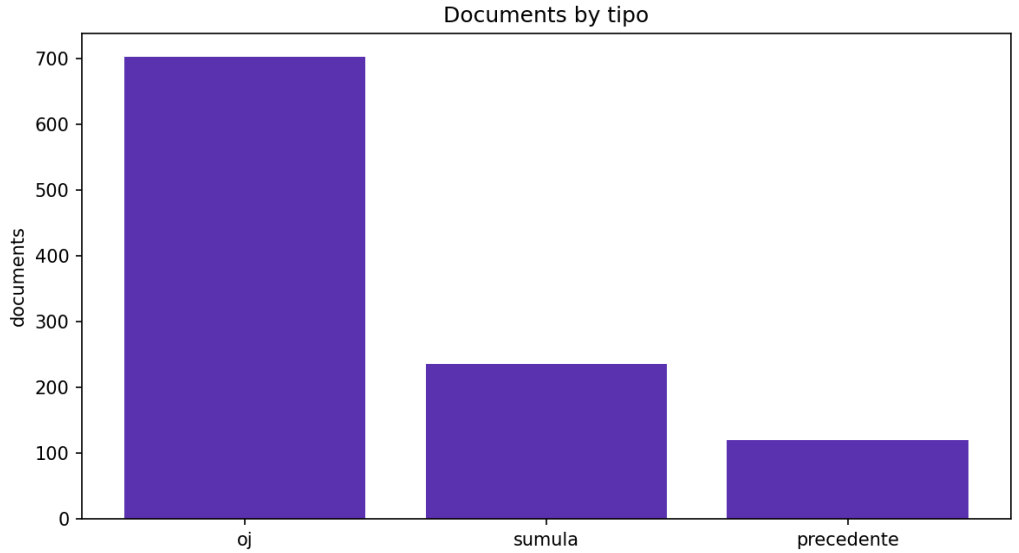
The project brief asks for a "document management system" with four parts:

Requirement	How we address it
Ingestion pipeline (CLI + API)	`capstone` CLI, **PDF and plain-text** input
Knowledge base	**ChromaDB** vector store + metadata
Exploration interface	**Streamlit + Plotly** dashboard, semantic search
Quality control	**Heuristic labels + supervised classifier + anomaly detection**

The domain is an **open-access official act**, so no copyrighted material is used.

Dataset

- Source: TST *Livro de Súmulas, OJs e Precedentes Normativos* (public act).
- **1058 verbetes** segmented from the 579-page PDF.



Bars = documents per type: **703 OJs**, **235 Súmulas**, **120 Precedentes**.

Concept: what is an "embedding"?

- An embedding is a **list of numbers (a vector)** that represents the *meaning* of a piece of text.
- Our model turns each verbete into **384 numbers**.
- Texts with similar meaning end up **close together** in this 384-dimensional space; unrelated texts end up far apart.

This is what lets us search by *meaning* rather than by exact keywords.

How we embed

- Model: `paraphrase-multilingual-MiniLM-L12-v2` (Sentence-Transformers).
- **Multilingual** (trained on many languages) so Portuguese legal text works without fine-tuning.
- **384 dimensions**, runs on **CPU**, batched (32 at a time).
- We embed `tema` (theme) + `text` together, and **normalise** the vectors.

Concept: cosine similarity and vector search

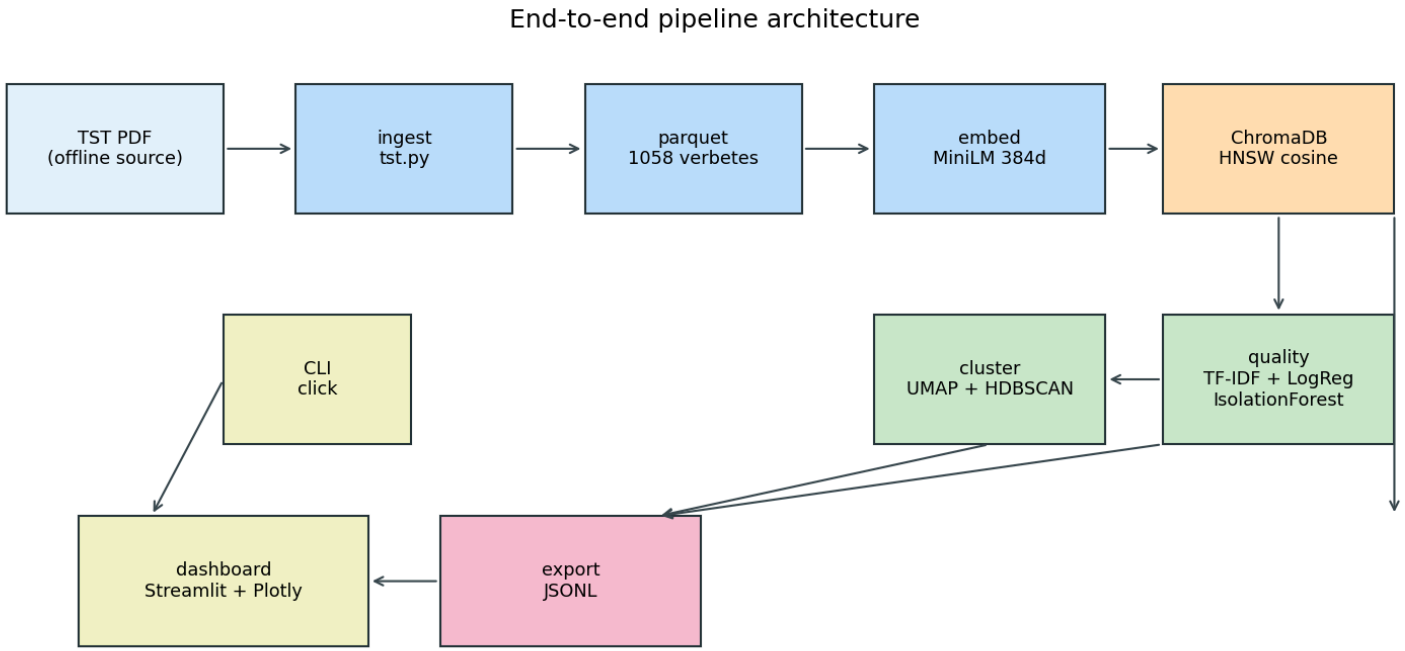
- **Cosine similarity** measures the **angle** between two vectors: `1.0` = same direction (very similar), `0.0` = unrelated.
- To search, we embed the query and return the verbetes whose vectors are closest to it.
- Because vectors are normalised, ranking by cosine similarity is a simple, fast operation.

This is the core of "semantic search": we compare meaning, not words.

Concept: vector database (ChromaDB)

- A normal database stores rows; a **vector database** stores vectors and finds the **nearest neighbours** to a query vector, fast.
- We use **ChromaDB**, persisted on disk, with an **HNSW** index (a graph that speeds up approximate nearest-neighbour search).
- Distance space: **cosine**. Each record keeps its **metadata** (`tipo`, `orgao`, `codigo`, `tema`, quality label, cluster, anomaly score) so we can **filter** as well as search.

System architecture




```
Ingest -> Embed -> ChromaDB -> Cluster -> Quality -> Export -> Interface
```

Pipeline stages

1. **Ingest** - segment PDF/plain text into verbetes, keep current wording.
2. **Embed** - 384-D multilingual vectors (theme + text).
3. **Store** - ChromaDB (HNSW, cosine) with metadata.
4. **Cluster** - UMAP projection + HDBSCAN.
5. **Quality** - heuristic labels, TF-IDF + logistic-regression classifier, IsolationForest anomalies.
6. **Export** - enriched JSONL that powers the dashboard.

Every stage runs alone *or* as one `capstone pipeline` command.

Ingestion

- Parse with `pdfplumber`; detect headings per verbete type.
- Cut at `Histórico:` and repeated headers, stop at the subject index.
- Keep only the **current wording** of each verbete.
- Files detected by extension: `.pdf` or `.txt` (one verbete per line).
- Attach **heuristic quality labels** during ingestion (next concept slides).

Live demo

Dashboard (`capstone-dashboard` or `make dashboard`):

1. **Search** tab - semantic search, filters by type and quality.
2. **Cluster map** tab - UMAP scatter, cluster sizes, top terms.
3. **Quality & anomalies** tab - label mix, length histogram, top anomalies.

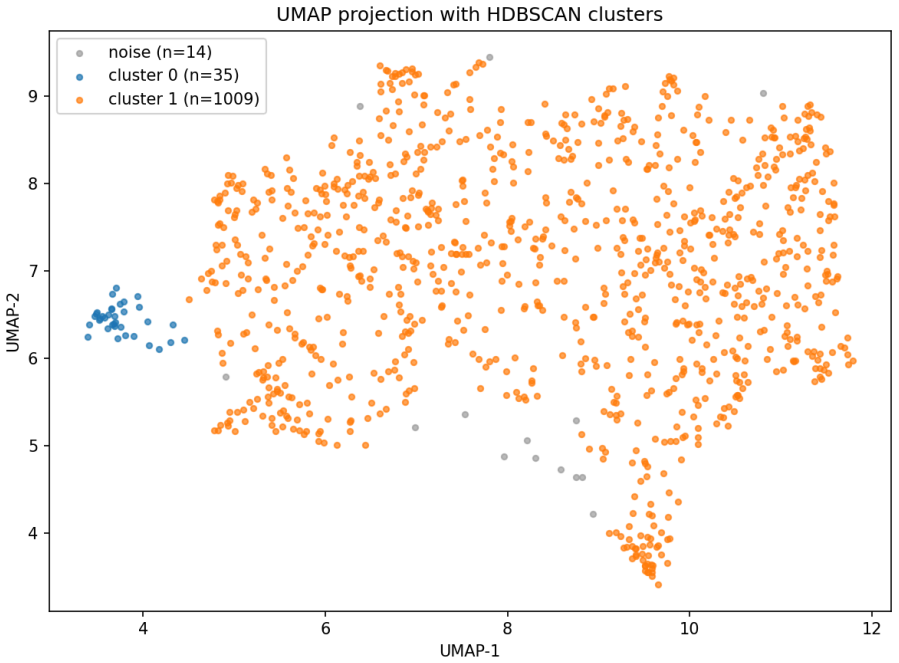
Same engine via CLI:

```
capstone search "adicional de insalubridade" -k 5
capstone clusters --top-terms 5
capstone anomalies --limit 10
capstone stats
```

Concept: UMAP (dimensionality reduction)

- We cannot plot 384 dimensions. **UMAP** flattens them to **2D** while keeping nearby points nearby.
- Two knobs:
  - `n_neighbors=15` - balances **local** structure vs **global** structure.
  - `min_dist=0.1` - how tightly points may pack.
- We also cluster on the **2D UMAP coordinates**.

Clustering results



UMAP (`n_neighbors=15`, `min_dist=0.1`, cosine) + HDBSCAN (`min_cluster_size=15`): **1009** main cluster, **35** historical, **14** noise.

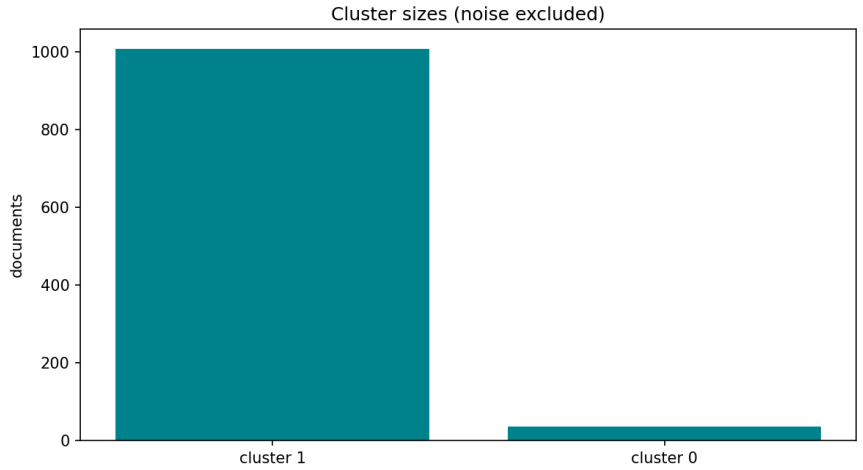
How to read: `x` / `y` = UMAP-1/UMAP-2 (no units); colour = cluster, grey = noise; legend shows sizes.

Concept: HDBSCAN (density-based clustering)

- Groups points that are **densely packed**, without being told how many clusters to expect.
- Handles **clusters of different sizes/densities** (unlike k-means).
- Points in sparse regions are labelled **noise (`-1`)** - honest about uncertainty.

Here most of the book shares one legal vocabulary (one big cluster), while a small group of **historical rural-work precedentes** separates cleanly.

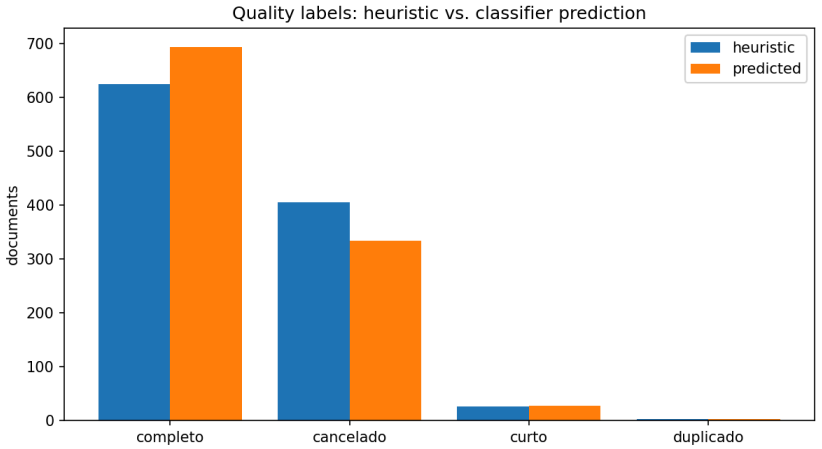
Cluster sizes



Bars = documents per cluster (noise excluded).

- Cluster 1: **1009** verbetes - the general body of the book.
- Cluster 0: **35** verbetes - historical rural-work homologation language.
- Noise: **14** verbetes not dense enough to belong anywhere.

Quality control: heuristic labels



How to read: blue = heuristic rule, orange = classifier prediction, per label.

Rules: cancelled/revoked keywords, duplicates, very short text, noise.

Concept: TF-IDF + logistic regression

- **TF-IDF** turns text into numbers: common words count little, rare and informative words count a lot. We use single words and pairs (1-2 grams), up to 5000 features.
- **Logistic regression** is a simple, fast linear classifier: it learns which words/pairs point to each quality label.
- `class_weight="balanced"` compensates for the rare classes.

This is **supervised learning** on labels we created with rules ("weak labels").

Classifier results

Trained on the heuristic labels, evaluated on a held-out **20%** split:

Class	Precision	Recall	F1	n
cancelado	0.99	0.84	0.91	81
completo	0.87	0.98	0.92	125
curto	0.00	0.00	0.00	5
duplicado	1.00	1.00	1.00	1

Accuracy 0.91, macro-F1 0.71, weighted-F1 0.90.

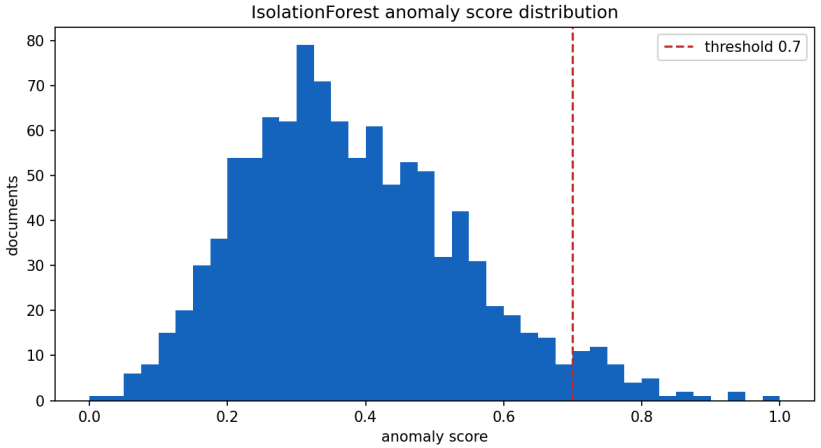
Honest note: the two tiny classes (5 and 1 examples) cannot be learned reliably.

Concept: IsolationForest (anomaly detection)

- Builds many random **trees** that split the data.
- **Anomalies are isolated in fewer splits**, so they get a **higher score**.
- `contamination=5%` sets the expected outlier rate; `n_estimators=200` trees.
- Scores are normalised to **0-1** (higher = more anomalous).

It is **unsupervised**: nobody told it what an "anomaly" looks like.

Anomalies found



How to read: `x` = score (0-1), `y` = documents, red dashed line = 0.7 threshold.

47 verbetes (4.4%) above 0.7; top hits are atypical historical precedentes.

Engineering quality

- **Tests:** 14 pytest cases, including an **offline end-to-end** pipeline run.
- **CI:** GitHub Actions (lint + strict typing) and GitLab CI (pipeline + tests).
- **Compliance:** `black` and `mypy` (strict) across the repository.
- **Docker:** `docker compose up --build` builds the pipeline, pre-loads the offline fixture and serves the dashboard - no network needed at runtime.

```
make test lint docs
docker compose up --build   # http://localhost:8501
```

Reproducibility

```
capstone pipeline -i data/raw/livrointernet12pdf.pdf
make docs        # figures + PDF reports
make dashboard   # interactive UI
```

- Offline fixture (`tests/fixtures/tst-sample.pdf`, 22 verbetes) used by CI and Docker.
- Machine-readable results: `docs/figures/summary.json`.

Limitations

- Quality labels are **heuristic**, not legally authoritative.
- The embedding model is **general-purpose**, not legal-domain fine-tuned.
- Tiny classes (`curto`, `duplicado`) are **not reliably learned**.
- Clustering is sensitive to density parameters and the 2D projection is lossy.
- Educational project - **no legal advice**.

Future work

- Domain-tuned or legal Portuguese embeddings.
- **Active learning:** human corrections feed the classifier.
- Evaluate retrieval quality against a labelled relevance set.
- Larger corpora and incremental ingestion.
- Richer entity extraction (themes, dates, revocation graphs).

Takeaways

- A complete pipeline from **raw PDF to interactive knowledge base**.
- Embeds, stores, visualises, clusters and **quality-checks** a real legal corpus.
- Reproducible, tested and containerised end to end.

Thanks - questions?

References (with a one-line explanation)

1. **TST** - *Livro de Súmulas, OJs e Precedentes Normativos*. The data source (official public act, not copyrighted).
2. **Reimers & Gurevych, 2019** - *Sentence-BERT*. Shows how to train transformer networks to produce good sentence embeddings (our embedding family).
3. **McInnes et al., 2018** - *UMAP*. The dimensionality-reduction method we use to visualise the vectors in 2D.
4. **Campello et al., 2013** - *HDBSCAN*. The density-based clustering method.
5. **Liu, Ting & Zhou, 2008** - *Isolation Forest*. The anomaly-detection method.

Backup: command reference

```
capstone ingest -i data/raw/livrointernet12pdf.pdf -o data/processed/tst.parquet
capstone embed --batch-size 32
capstone cluster
capstone quality --contamination 0.05
capstone export -o data/final/tst.jsonl
capstone search "hora extra" -k 10
```

Backup: glossary

- **Embedding** - numeric vector for the meaning of a text.
- **Cosine similarity** - angle-based similarity between vectors (1 = same).
- **Vector DB / HNSW** - store and fast-nearest-neighbour search over vectors.
- **UMAP** - reduces many dimensions to 2 for visualisation.
- **HDBSCAN** - density-based clustering; marks sparse points as noise.
- **TF-IDF** - numeric text features weighting rare/informative words.
- **Logistic regression** - simple linear classifier.
- **IsolationForest** - unsupervised anomaly detector.
- **Transfer learning** - reusing a model pre-trained on a large general task.